Suitable Visualization Techniques for Evaluating a Classification Model

Evaluating a classification model should not depend only on a single metric such as **accuracy**. A model may have high accuracy but still perform poorly for certain classes, especially when the dataset is imbalanced.

Visualization techniques help us understand:

- Which classes are correctly classified
- Which classes are confused with each other
- How well the model separates different classes
- The trade-off between correct and incorrect predictions

The main visualization techniques used for evaluating classification models are:

1. **Confusion Matrix**
2. **ROC Curve and AUC**
3. **Precision-Recall Curve**
4. **Classification Metric Comparison Charts**

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1. Confusion Matrix

A **Confusion Matrix** is one of the most important tools for evaluating a classification model.

It compares:

- **Actual class labels**
- **Predicted class labels**

For a binary classification problem, it contains four possible outcomes.

Actual / Predicted	**Positive**	**Negative**
Positive	True Positive (TP)	False Negative (FN)
Negative	False Positive (FP)	True Negative (TN)

Meaning of the Terms

True Positive (TP)

The model correctly predicts the positive class.

Example: A spam email is correctly classified as spam.

True Negative (TN)

The model correctly predicts the negative class.

Example: A normal email is correctly classified as not spam.

False Positive (FP)

The model incorrectly predicts a negative case as positive.

Example: A normal email is incorrectly classified as spam.

This is also called a **False Alarm**.

False Negative (FN)

The model incorrectly predicts a positive case as negative.

Example: A spam email is incorrectly classified as normal.

Why the Confusion Matrix is Useful

A confusion matrix shows **where the model makes mistakes**.

For example, in multi-class classification, it can show whether:

- Cats are confused with dogs.
- Certain digits are confused with similar digits.
- One class is consistently misclassified.

Therefore, it provides more detailed information than accuracy alone.

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2. ROC Curve (Receiver Operating Characteristic Curve)

The **ROC Curve** evaluates how well a binary classifier separates two classes at different classification thresholds.

It plots:

- **X-axis:** False Positive Rate (FPR)
- **Y-axis:** True Positive Rate (TPR)

True Positive Rate

$$TPR = \frac{TP}{TP + FN}$$

This is also called **Recall** or **Sensitivity**.

False Positive Rate

$$FPR = \frac{FP}{FP + TN}$$

Interpretation of ROC Curve

A good classifier produces a curve that moves toward the **top-left corner**.

True Positive Rate

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| ●———

| /

| /

| /

|/_____ False Positive Rate

The diagonal line represents the performance of a **random classifier**.

A curve closer to the top-left corner indicates better classification performance.

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3. AUC (Area Under the ROC Curve)

The **AUC** represents the area under the ROC curve.

Its value ranges from:

$$0 \text{ to } 1$$

Interpretation

AUC Value	Interpretation
1.0	Perfect classifier
0.9 – 1.0	Excellent
0.8 – 0.9	Good
0.7 – 0.8	Acceptable
0.5	Random classifier

A larger AUC indicates that the model has a better ability to distinguish between classes.

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4. Precision-Recall Curve

The **Precision-Recall (PR) Curve** is especially useful when the dataset is **imbalanced**.

For example:

- Fraud detection
- Disease detection
- Spam detection

In such problems, accuracy can be misleading because one class may occur much more frequently than the other.

Precision

Precision measures how many predicted positive cases are actually positive.

$$Precision = \frac{TP}{TP + FP}$$

Recall

Recall measures how many actual positive cases are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

Interpretation

A good classification model should achieve:

- High precision
- High recall

The Precision-Recall Curve shows the trade-off between these two measures as the classification threshold changes.

It is particularly useful when detecting the **positive class is more important**.

Illustrative Example: Email Spam Classification

Consider a classification model that predicts whether an email is:

- **Spam**
- **Not Spam**

Suppose the model is tested on **100 emails**.

The following confusion matrix is obtained:

Actual / Predicted	**Predicted Spam**	**Predicted Not Spam**
Actual Spam	45	5
Actual Not Spam	8	42

Therefore:

$$TP = 45$$
$$FN = 5$$
$$FP = 8$$
$$TN = 42$$

A. Confusion Matrix Interpretation

The model correctly identifies:

- **45 spam emails**
- **42 normal emails**

However:

- **5 spam emails** are incorrectly classified as normal.
- **8 normal emails** are incorrectly classified as spam.

The confusion matrix immediately shows the types of errors made by the classifier.

B. Classification Metrics

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$
$$Accuracy = \frac{45 + 42}{100}$$
$$\boxed{Accuracy = 87\%}$$

Precision

$$Precision = \frac{45}{45 + 8}$$
$$\boxed{Precision \approx 84.9\%}$$

This means that when the model predicts an email as spam, approximately **84.9% of those predictions are actually spam**.

Recall

$$Recall = \frac{45}{45 + 5}$$
$$\boxed{Recall = 90\%}$$

This means the model correctly identifies **90% of all actual spam emails**.

F1-Score

The F1-score combines Precision and Recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

For this example:

$$\boxed{F1 \approx 87.4\%}$$

Visualizing the Example

A confusion matrix visualization would clearly show that most predictions occur along the **main diagonal**, representing correct classifications.

	Predicted	
	Spam	Not Spam
Actual Spam	45	5
Actual Not Spam	8	42

The diagonal values:

$$45 + 42$$

represent correct predictions.

The off-diagonal values:

$$5 + 8$$

represent classification errors.

ROC Curve Interpretation for the Example

Suppose the spam classifier is tested using different probability thresholds.

For example:

- At threshold 0.3, the model identifies more spam but may create more false positives.
- At threshold 0.5, there may be a balance between false positives and false negatives.
- At threshold 0.8, the model predicts spam only when highly confident, possibly missing some spam emails.

The ROC curve shows how the **True Positive Rate and False Positive Rate change across these thresholds**.

A model with an ROC curve close to the top-left corner and a high AUC value has better ability to distinguish between spam and normal emails.

Comparison of Classification Visualizations

Visualization	Main Purpose	What Indicates Good Performance
Confusion Matrix	Shows correct and incorrect classifications	High values on the main diagonal
ROC Curve	Evaluates class separation at different thresholds	Curve close to top-left corner
AUC	Summarizes ROC performance	Value close to 1
Precision-Recall Curve	Evaluates positive-class detection	High precision and recall
Metric Comparison Chart	Compares accuracy, precision, recall and F1-score	High values across important metrics

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Important Note: Accuracy Alone Can Be Misleading

Consider a disease detection dataset containing:

- 990 healthy patients
- 10 patients with the disease

A model that predicts **"Healthy" for every patient** would have:

$$Accuracy = 99\%$$

However, it would detect:

$$0\% \text{ of diseased patients}$$

Therefore, accuracy alone does not always provide a complete evaluation.

In such cases:

- Confusion Matrix
- Precision
- Recall
- F1-score
- Precision-Recall Curve

provide a much better understanding of model performance.

Final Exam Answer

Classification models can be evaluated using several visualization techniques because a single metric such as accuracy may not reveal all types of prediction errors. A **Confusion Matrix** displays the number of true positives, true negatives, false positives, and false negatives, helping identify which classes are correctly or incorrectly classified. The **ROC Curve** plots the True Positive Rate against the False Positive Rate at different classification thresholds, while the **AUC** summarizes the classifier's ability to distinguish between classes. The **Precision-Recall Curve** is particularly useful for imbalanced datasets because it focuses on the trade-off between correctly identifying positive cases and avoiding false positives.

For example, in a spam email classification model, a confusion matrix can show how many spam and non-spam emails are correctly classified. ROC and Precision-Recall curves can further show how the model's performance changes with different decision thresholds. Therefore, visualization techniques provide a detailed and reliable evaluation of a classification model and help identify weaknesses that cannot be detected using accuracy alone.

Quick memory trick: C-R-P

- **C** → Confusion Matrix
- **R** → ROC Curve and AUC
- **P** → Precision-Recall Curve